

User Manual

SignBridge – Sign Language Detection and Translation

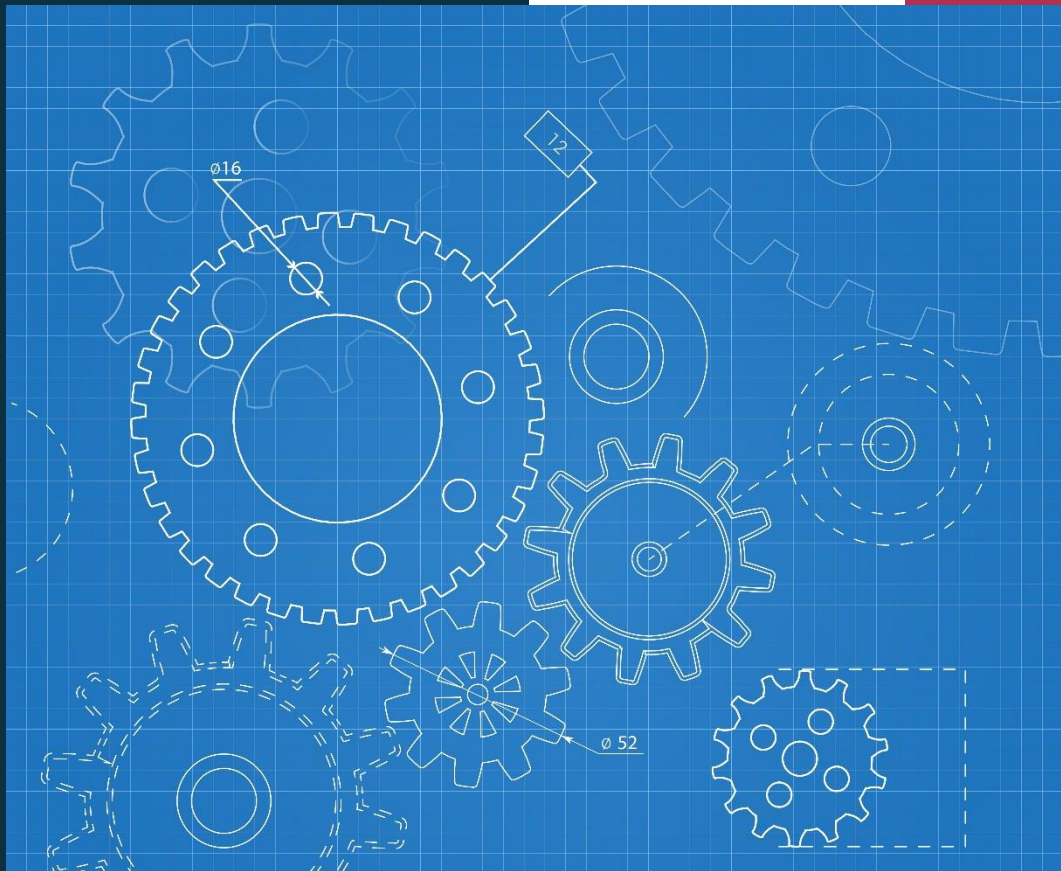


Table of Contents

1.0 Introduction	3
1.1 Key Features	3
1.2 Overview of How it Works	4
2.0 System Requirements.....	4
2.1 Hardware Requirements.....	4
2.2 Software and Browser Requirements.....	4
2.3 Internet Requirements	5
2.4 Environmental Requirements.....	5
2.5 Supported Platforms.....	5
3.0 Getting Started	6
3.1 Creating an Account.....	6
3.2 Logging In.....	8
3.3 Navigating Dashboard	9
3.4 Viewing Your Profile.....	10
3.5 Editing Your Profile	11
3.6 Resetting Password	12
3.7 Tips.....	13
3.8 Logging Out	13
4.0 Step-by-Step Usage Guide	15
4.1 Creating a Session.....	15
4.2 Joining a Session.....	16
4.3 Waiting Room	17
4.4 Using Gesture Recognition	19
4.5 Transcript	21
4.6 Using the Chat.....	22
4.7 Ending a Call.....	23
5.0 Help and Support Features.....	24
5.1 Help Section.....	24
5.2 Video Demo	24
5.3 User Guide	25
5.4 SLSL Reference Chart	25

5.5 Frequently Asked Questions	26
5.6 Contact Us.....	26
6.0 Error Handling/Troubleshooting.....	28
6.1 Camera Not Working	28
6.2 Gesture Not Recognised	28
6.3 Cannot Join a Call	28
6.4 Poor Video Quality/Lag	28
7.0 Current Limitations	28
7.1 Recognition Scope	29
7.2 Environmental Factors	29
7.3 Technical Limitations.....	29
7.4 Privacy Limitations	29
8.0 Privacy and Data Handling	30

1.0 Introduction

SignBridge is a web-based video conferencing platform designed to bridge the communication gap between Deaf and hearing individuals. The application uses the computer's webcam to recognise Sri Lankan Sign Language (SLSL) fingerspelling gestures in real time and translates them into text that appears on the dashboard of the screen.

Unlike standard video call applications, SignBridge is specifically built for conversations where one person uses sign language and the other does not. The hearing participant is able to read the translated text while the Deaf participant signs. This enables for natural, smooth, real-time communication without the need for an interpreter.

All video processing occurs locally on the device. The video feed is never uploaded to any server, ensuring privacy and security.

No technical background is required to use this application. This guide provides step-by-step instructions with screenshots to help you get started.

1.1 Key Features

Video Calls

- One-to-one video calls with standard controls.

Sign Language Recognition

- Real-time detection of Sri Lankan Sign Language fingerspelling gestures via webcam.

Text Translation

- Recognised signs are displayed as text on the screen.

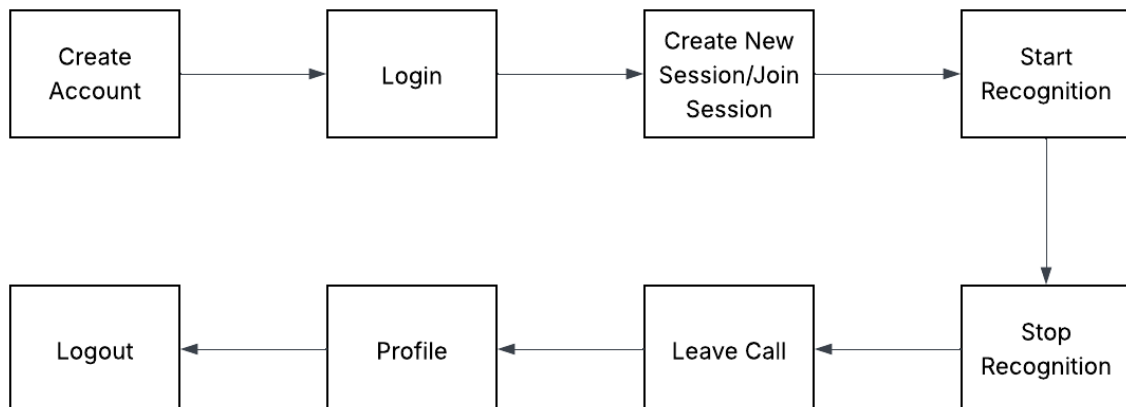
Running Transcript

- A log of all recognised letters is maintained during the call.

No Installation Required

- The application works entirely within the web browser

1.2 Overview of How it Works



1. The Deaf user signs a letter from the Sign Language alphabet.
2. The webcam captures the gesture.
3. The system detects the hand and identifies the sign.
4. The corresponding letter appears as text on screen.
5. The hearing user reads the text and can respond via the chat.

2.0 System Requirements

2.1 Hardware Requirements

Component	Minimum Requirements
Webcam	720p
Processor	Intel i5 (8 th generation), Ryzen 5, M1
RAM	8 GB

2.2 Software and Browser Requirements

This application works entirely in your web browser. It is recommended to use the latest version of your browser.

Browser	Version
Google Chrome	Latest version
Mozilla Firefox	Latest version
Microsoft Edge	Latest version
Safari	Latest version

2.3 Internet Requirements

Metric	Minimum	Recommended
Upload Speed	5 Mbps	10 Mbps
Download Speed	5 Mbps	10 Mbps
Latency	<150 ms	<50 ms
Connection	Stable broadband	Fibre

2.4 Environmental Requirements

Factor	Recommendation	Reason
Lighting	Well-lit room with light facing your face, not behind you.	Poor lighting makes it harder for the system to detect hand gestures.
Background	Plain, uncluttered background.	Reduces the interference with hand detection.
Camera Position	Camera should be eye level with your hands visible in the frame.	Ensures gestures are captured properly.
Distance from Camera	Sit 1/2 feet from the camera.	Allows the system to see your hand clearly.
Clothing	Avoid long sleeves or jewellery that may cover your hands/fingers.	Keeps hand shapes fully visible.

2.5 Supported Platforms

- Windows 10/11
- MacOS
- Linux
- Android

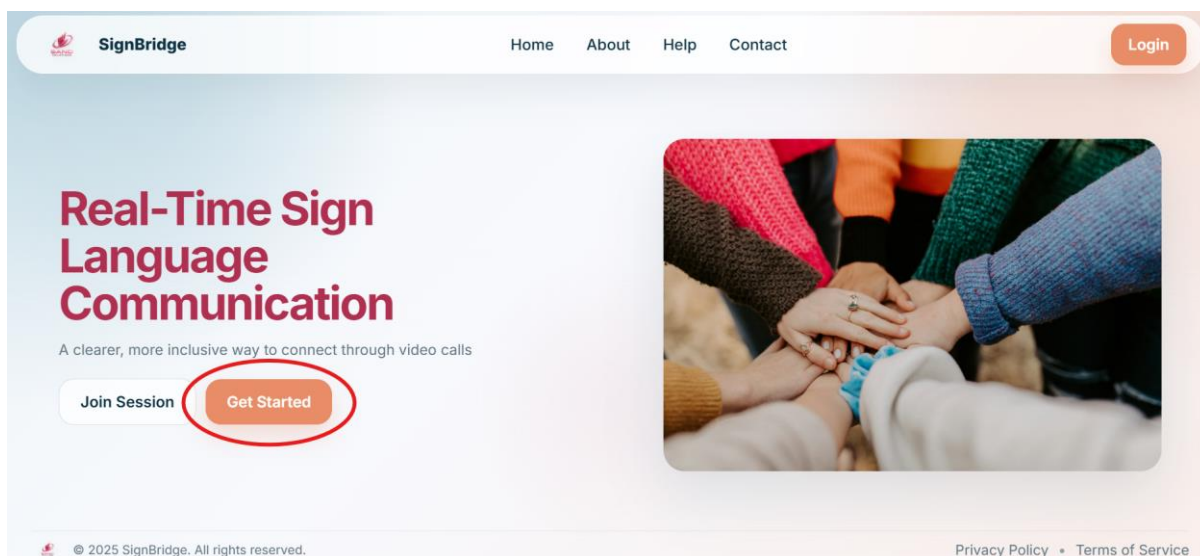
3.0 Getting Started

This section guides you through creating an account, logging in and navigating the dashboard.

3.1 Creating an Account

Follow the steps below to create an account.

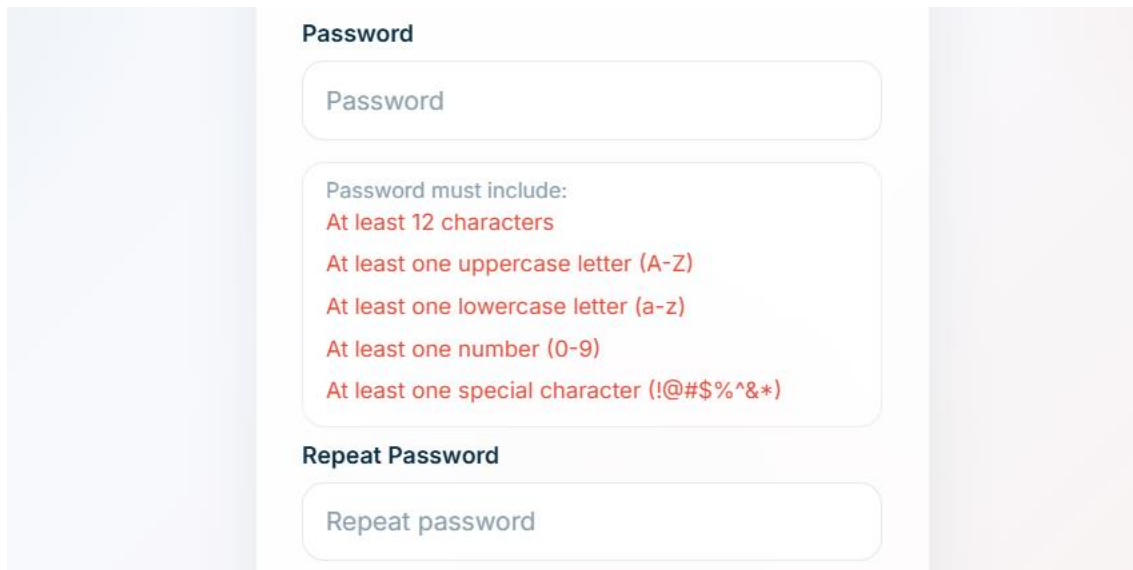
1. On the homepage, click on “Get Started”.



2. Fill in the fields with your email address, username and password.

A screenshot of the SignBridge "Create Account" form. The form is titled "Create Account" and includes the instruction "Set up your account to start using SignBridge." It contains the following fields: "Email" (with a placeholder "E-mail"), "Username" (with a placeholder "Username"), "Password" (with a placeholder "Password"), and "Repeat Password" (with a placeholder "Repeat password"). Below these fields is a checkbox labeled "I'm not a robot" next to a reCAPTCHA logo. At the bottom of the form is a "Sign Up" button. The form is set against the same homepage background as the previous image.

3. Make sure to create a password that fulfils the requirements in red.



Password

Password

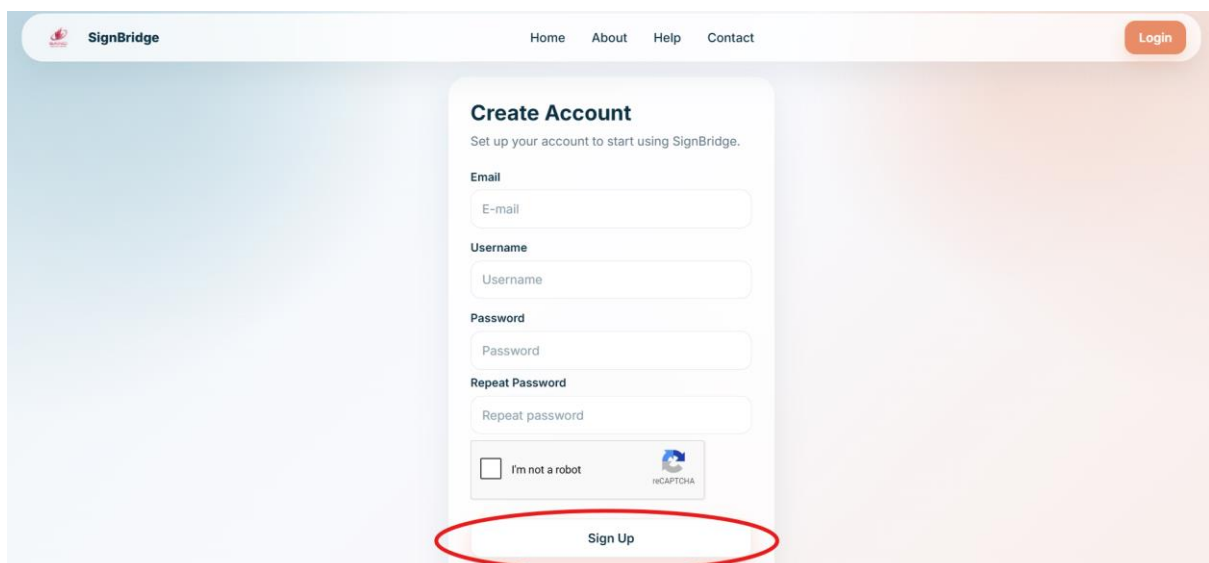
Password must include:

- At least 12 characters
- At least one uppercase letter (A-Z)
- At least one lowercase letter (a-z)
- At least one number (0-9)
- At least one special character (!@#\$%^&*)

Repeat Password

Repeat password

4. Check the reCAPTCHA box.
5. Click “Sign Up”.



SignBridge Home About Help Contact Login

Create Account

Set up your account to start using SignBridge.

Email

E-mail

Username

Username

Password

Password

Repeat Password

Repeat password

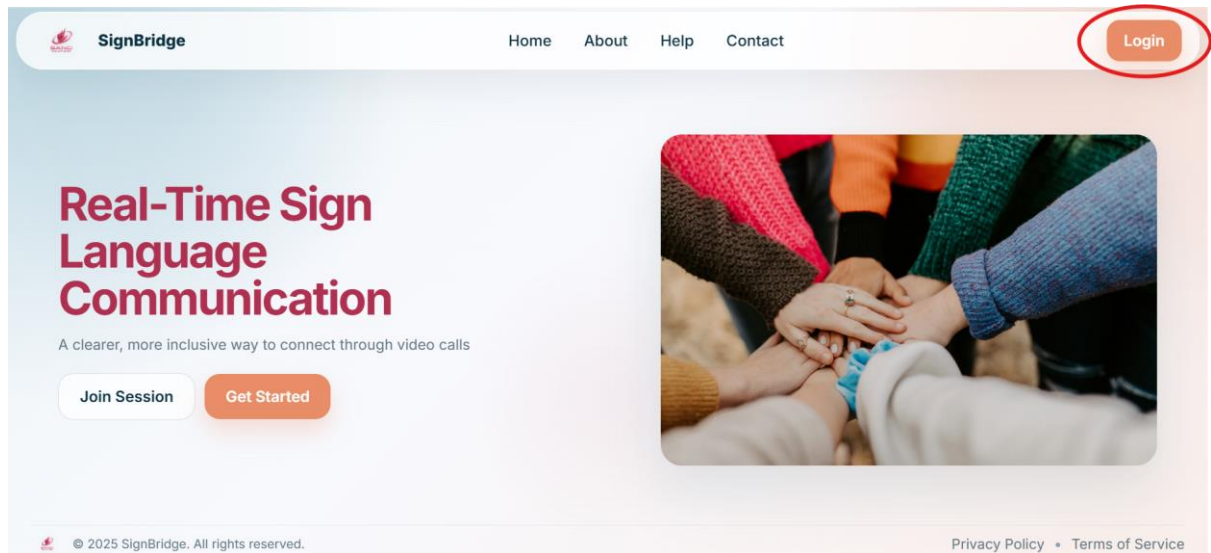
☐ I'm not a robot 

Sign Up

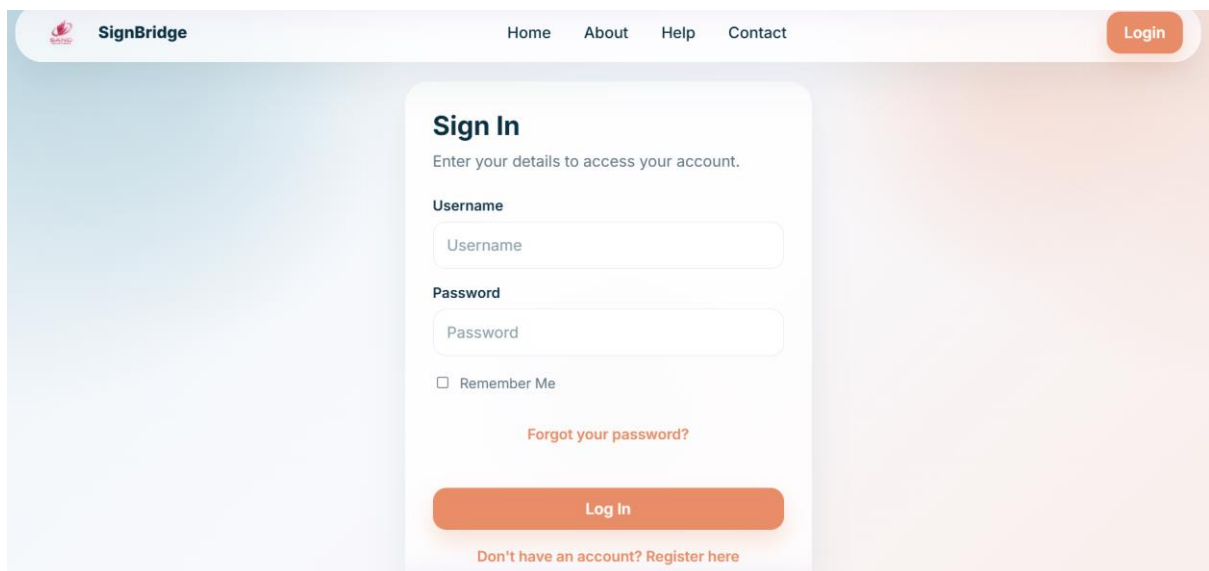
3.2 Logging In

Once you have created an account, follow the steps below to log in.

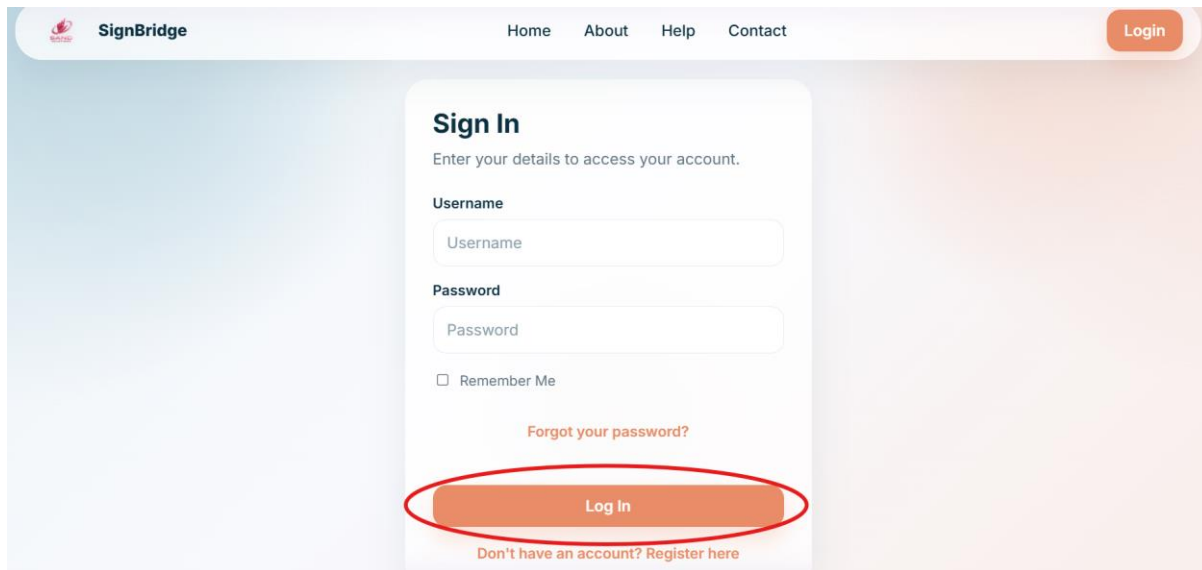
1. Click on the “Login” button on the homepage.



2. Fill in the username and password fields.

A screenshot of the Sign In form on the SignBridge website. The form is titled "Sign In" and includes the instruction "Enter your details to access your account." It contains two input fields: "Username" and "Password". Below these fields is a checkbox labeled "Remember Me". A link "Forgot your password?" is positioned below the checkbox. At the bottom of the form is a red "Log In" button. Below the button is a link "Don't have an account? Register here". The background of the page shows the same header and navigation links as the previous screenshot.

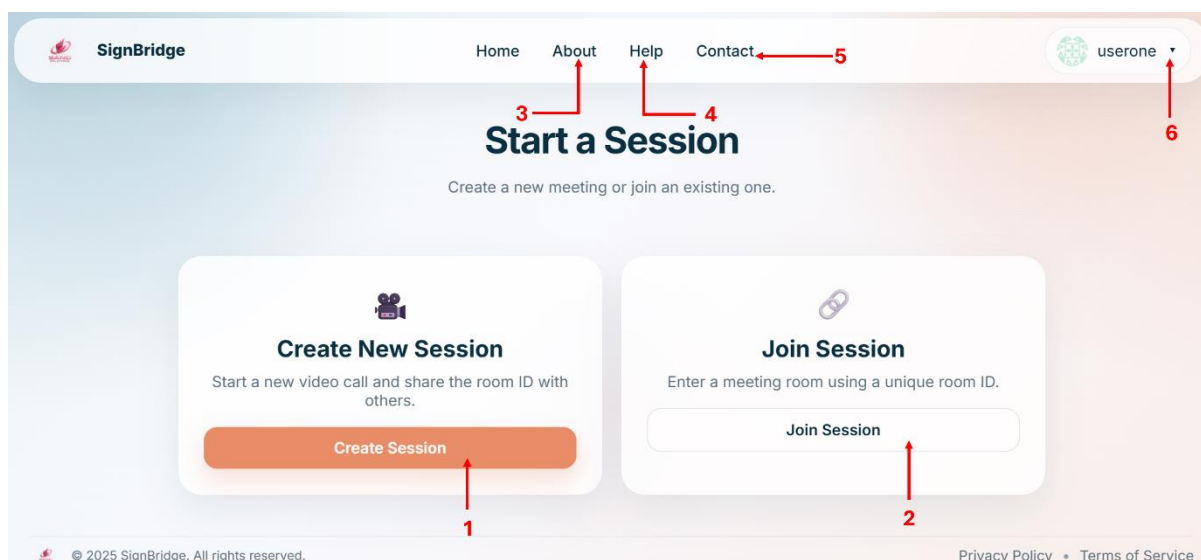
2. Click the “Log In” button.



The image shows the SignBridge Sign In page. At the top, there is a navigation bar with the SignBridge logo, links for Home, About, Help, and Contact, and a Login button. The main content area features a Sign In form with fields for Username and Password, a Remember Me checkbox, and a Forgot your password? link. The Log In button is highlighted with a red circle. Below the button is a link for users who don't have an account.

3.3 Navigating Dashboard

Once you have logged in, you will see the Dashboard which is the main hub for accessing the application’s features.

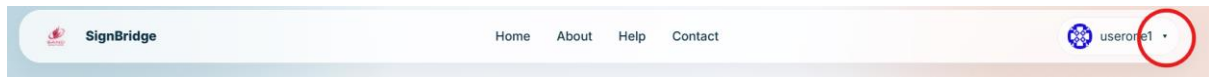


1. Create a new video call session.
2. Join an active video call you have been invited to.
3. Provides information about SignBridge as a project and the team building it.
4. Support page with instructions and tutorials on navigating the website.
5. Contact page if you have further questions or need support.
6. Drop down menu to your personal profile.

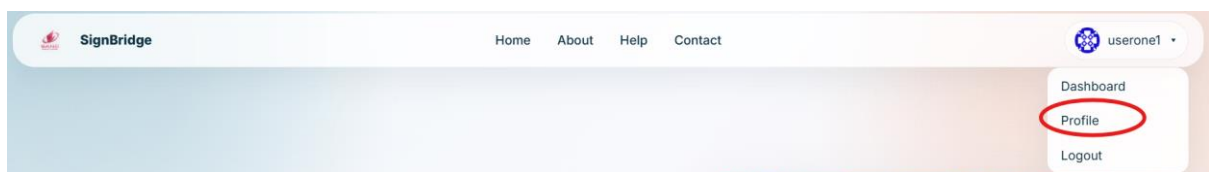
3.4 Viewing Your Profile

The profile section allows you to view your account information.

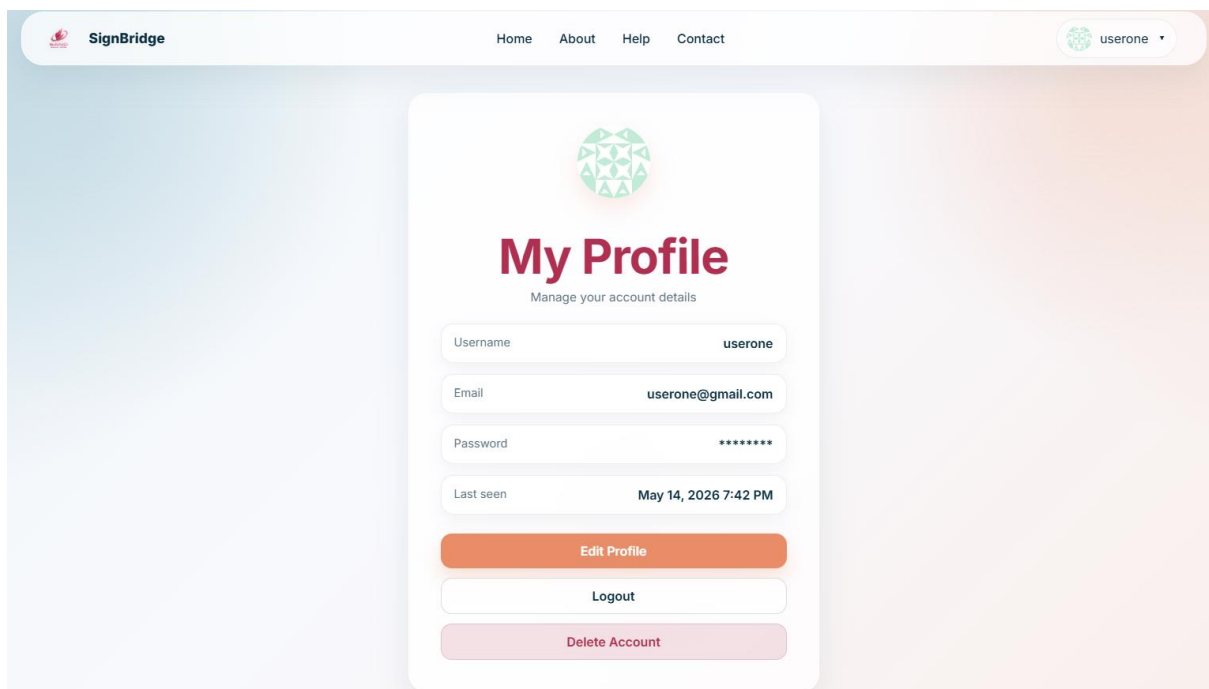
1. Click on the dropdown menu next to your username.



2. Click "Profile".



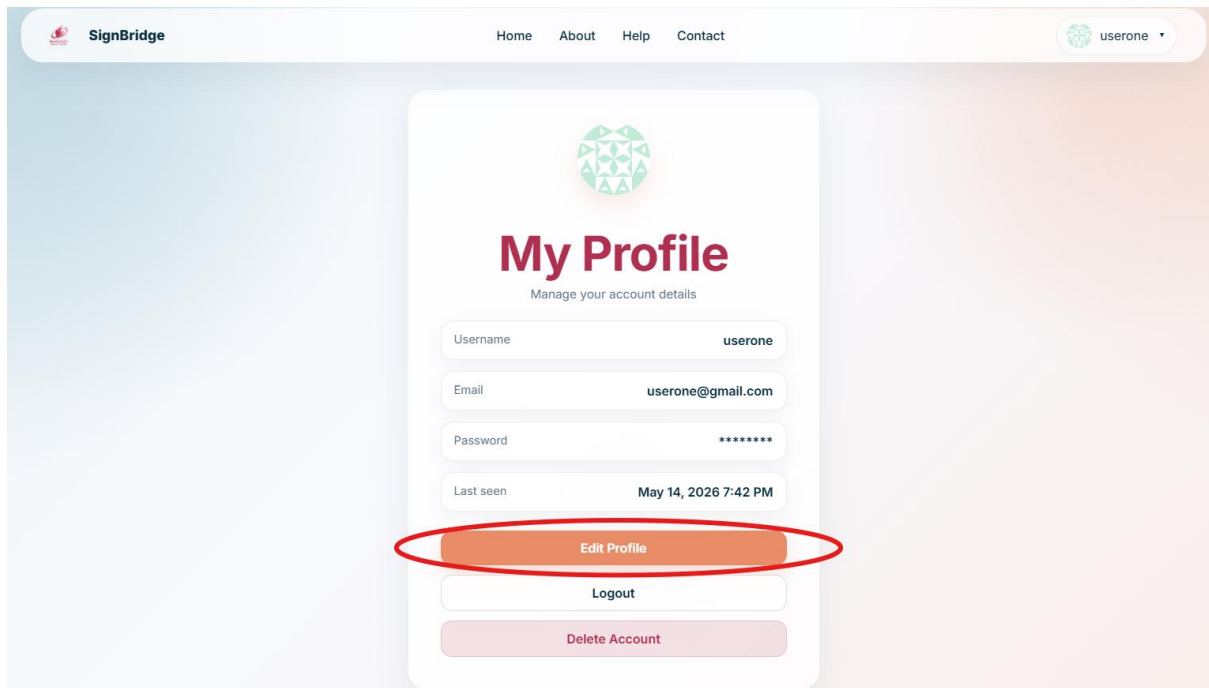
3. Your username and email address will be displayed.



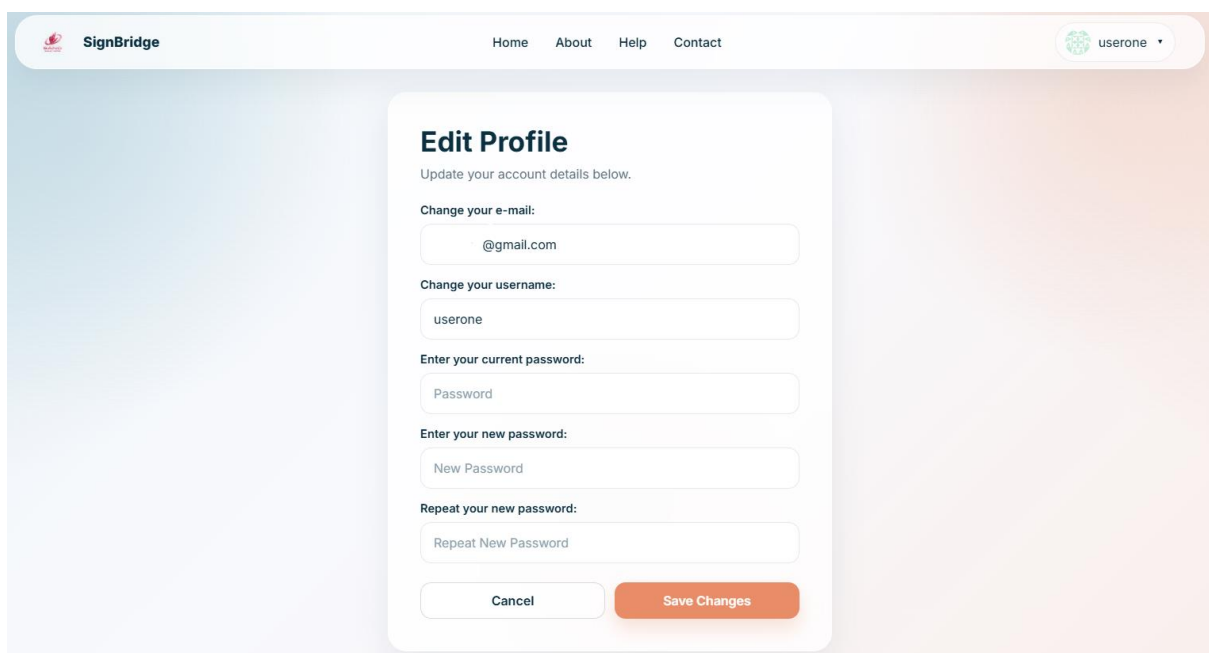
3.5 Editing Your Profile

Users can edit their personal information through the profile page.

1. Navigate to the profile page.
2. Click the “Edit Profile” button.



3. Update the information that needs to be changed.

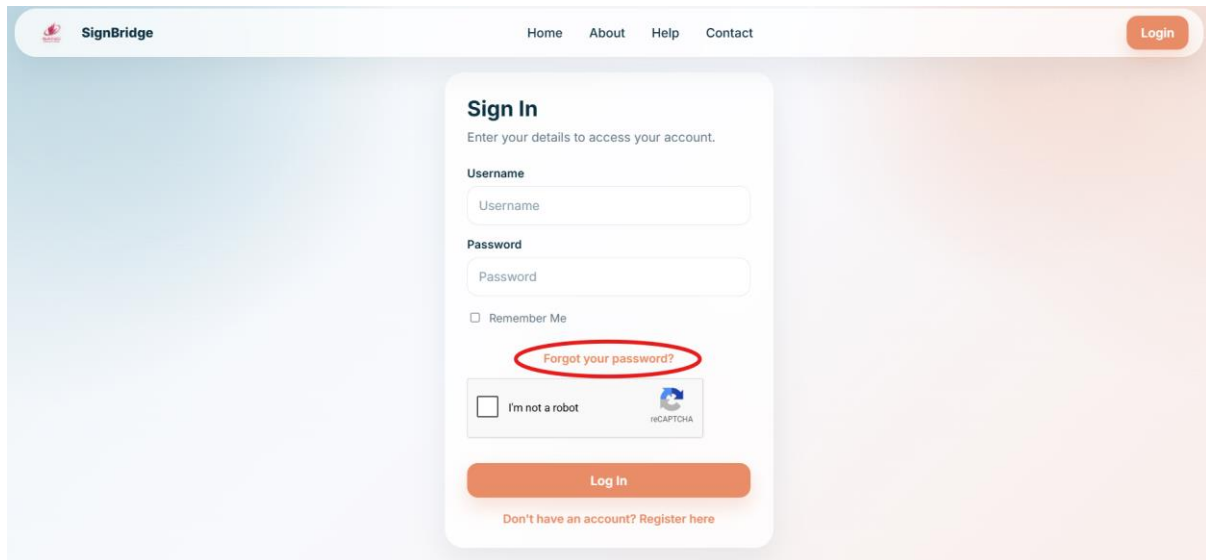


4. Click “Save changes.”

3.6 Resetting Password

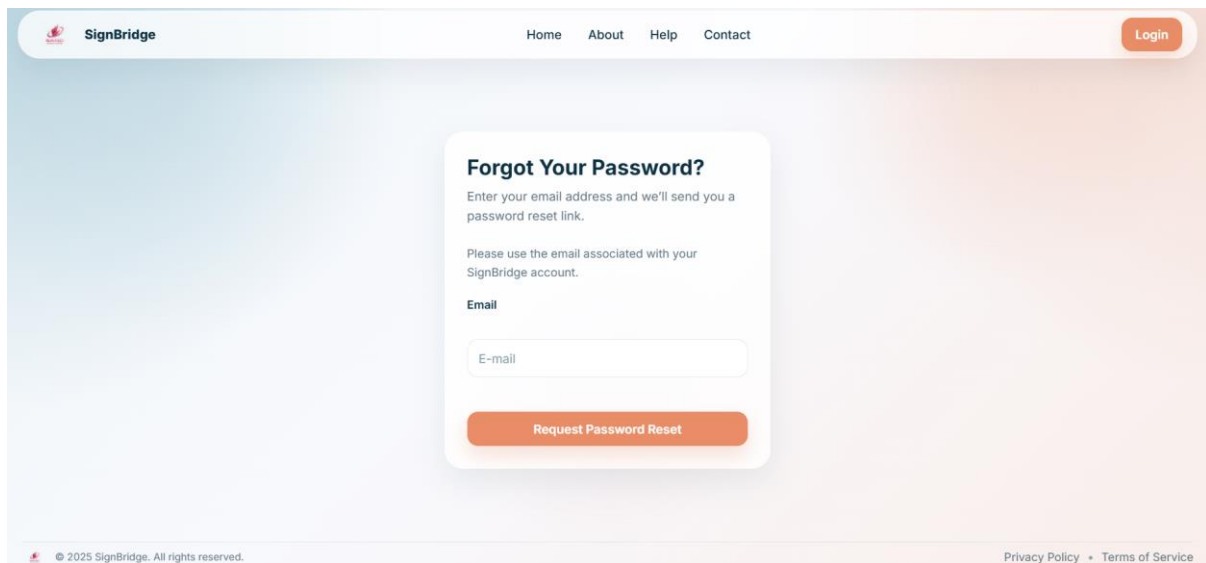
If you forget or want to reset your password, follow the steps below to reset it.

1. On the Sign In page, click “Forgot your password?”



The image shows the Sign In page of the SignBridge application. The page has a header with the SignBridge logo, navigation links (Home, About, Help, Contact), and a Login button. The main content area contains a Sign In form with fields for Username and Password, a Remember Me checkbox, and a link for "Forgot your password?" which is circled in red. Below the form is a Log In button and a link for "Don't have an account? Register here".

2. Enter the email address that's associated with your SignBridge account.



The image shows the "Forgot Your Password?" page of the SignBridge application. The page has a header with the SignBridge logo, navigation links (Home, About, Help, Contact), and a Login button. The main content area contains a form titled "Forgot Your Password?" with instructions to enter an email address to receive a password reset link. There is an "Email" input field and a "Request Password Reset" button. The footer includes copyright information and links to Privacy Policy and Terms of Service.

3. Click “Request Password Reset”.
4. Follow the instructions sent to your email.
5. Enter the new password and click “Request Password Reset”.

The screenshot shows the SignBridge 'Reset Password' interface. At the top, there's a navigation bar with 'Home', 'About', 'Help', 'Contact', and a 'Login' button. The main content area features a white card with the title 'Reset Password' and the instruction 'Create a new password for your account.' Below this, there are two input fields: 'Password' and 'Repeat Password'. An orange button labeled 'Request Password Reset' is positioned below the input fields, with a link 'Return to Sign In' underneath it.

6. Login using your updated password.

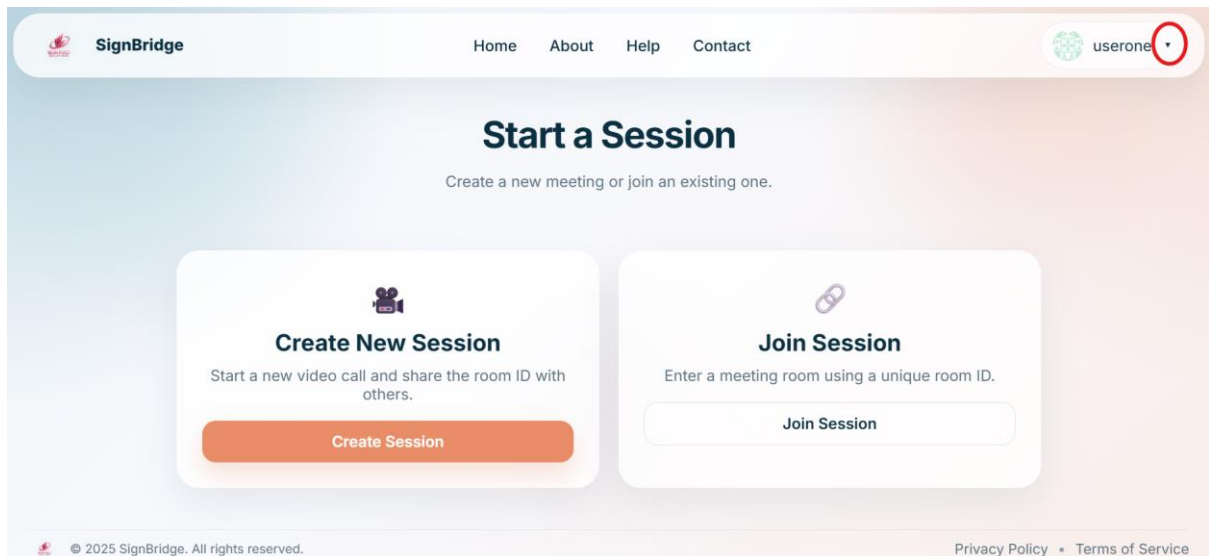
3.7 Tips

Tip	Explanation
Allow Camera Access	When prompted, click “Allow” to grant the application access to your webcam. The gesture recognition will not work without this.
Position Your Camera	Place the camera at eye level so your hands are clearly visible.
Setup Testing	Use the waiting room to preview your video before starting a call.
Review the SLSL Chart	Familiarise yourself with the chart in the Help section.

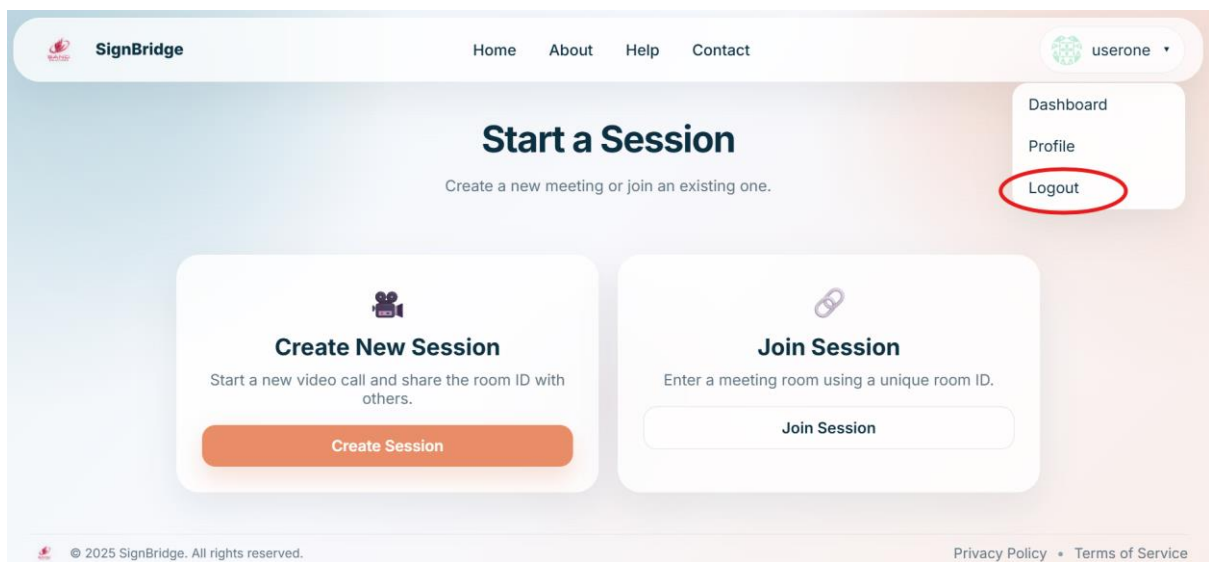
3.8 Logging Out

Follow the steps below to log out of your account.

1. Click on the drop down menu next to your username.



2. Click on the “Logout” button.



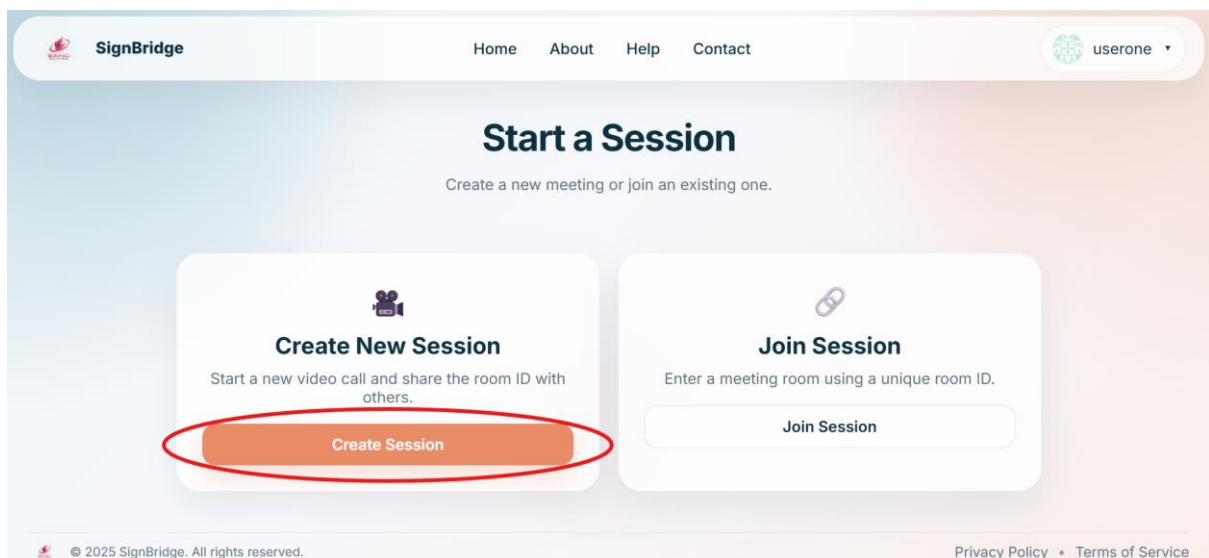
4.0 Step-by-Step Usage Guide

This section walks you through everything you need to know to use SignBridge. Each feature is explained with clear step-by-step instructions.

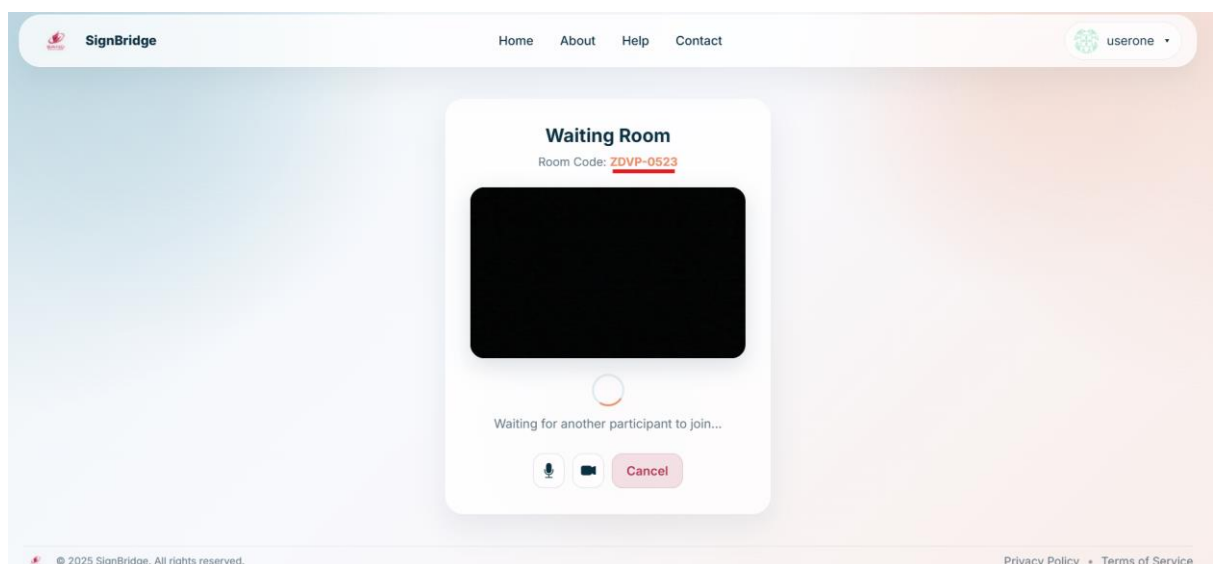
4.1 Creating a Session

Before starting a call, you need to create a session. This generates a unique Room ID that you will share with the other participant.

1. Click the “Create Session” button.



2. You will be taken to the waiting room.
3. The system will generate a unique Room ID.



4. Copy the Room ID
5. Share the Room ID with the person you want to call.

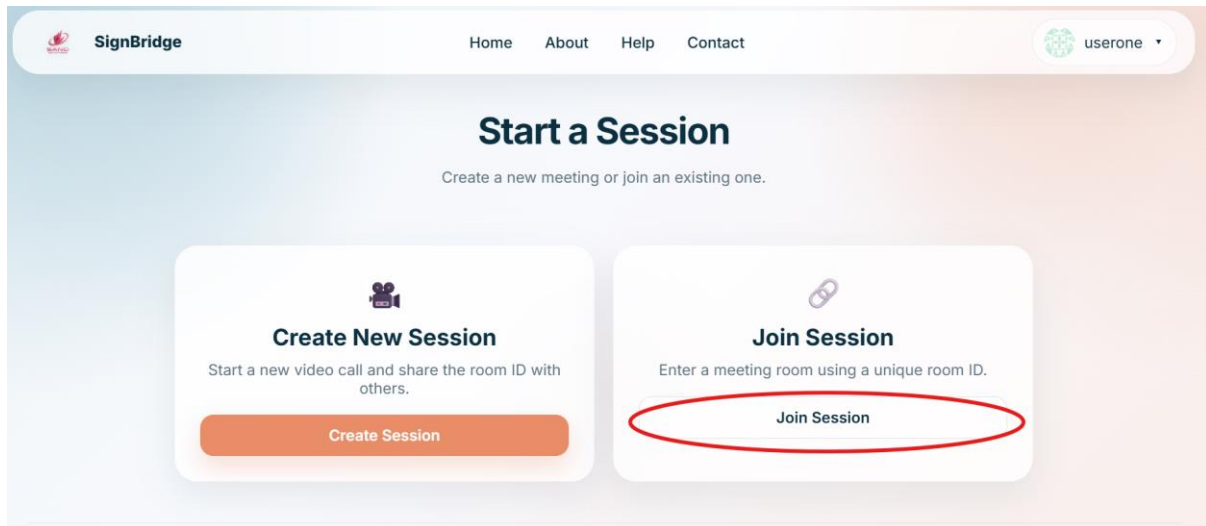
6. Once the other participant joins, the page will lead you to the video interface.

Note: Room IDs are temporary and expire after the call ends. A new ID is generated for each call.

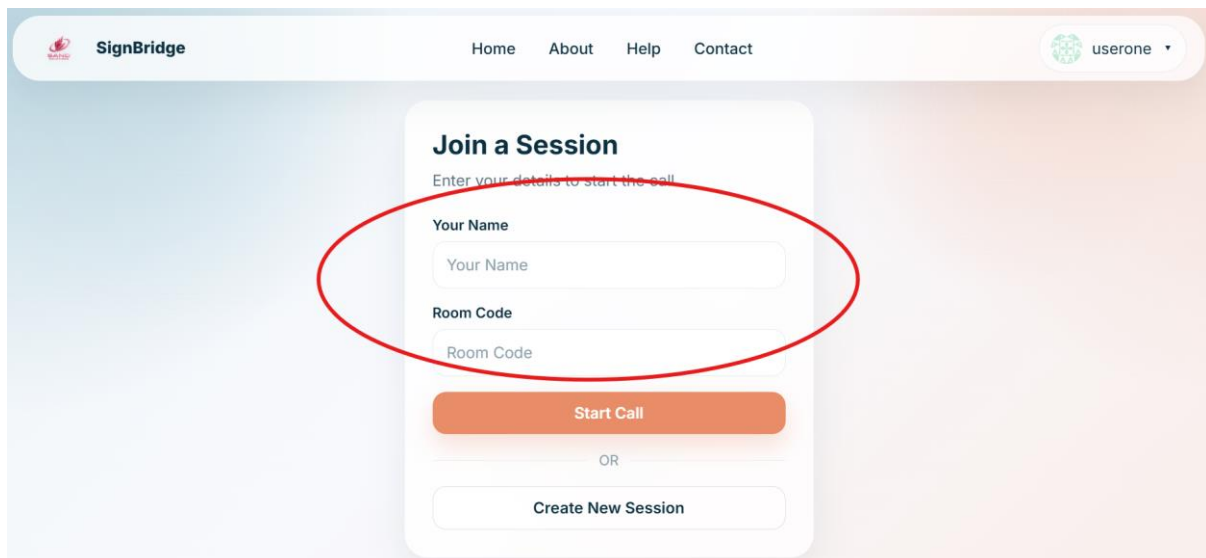
4.2 Joining a Session

If you have been shared a Room ID, follow the steps below to join the call.

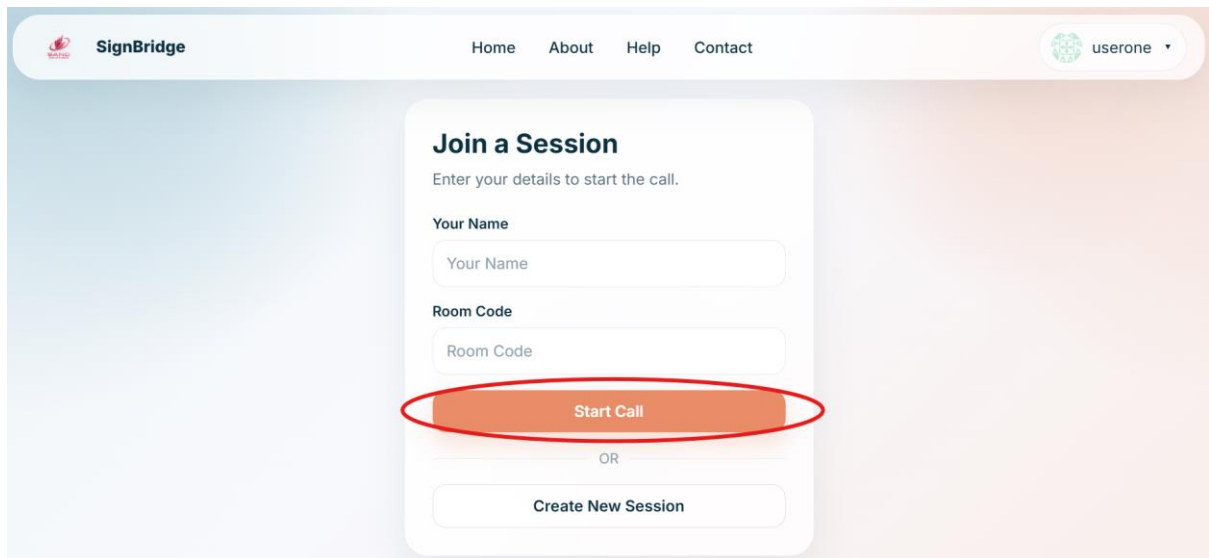
1. Click “Join Session” from the dashboard after logging in.



2. Enter your name and Room Code in the text field.



3. Click the “Start Call” button.



The image shows the 'Join a Session' form on the SignBridge website. The form is centered on a light blue and orange gradient background. It has a header 'Join a Session' and a sub-header 'Enter your details to start the call.' Below this are two input fields: 'Your Name' and 'Room Code'. A red oval highlights the 'Start Call' button, which is orange with white text. Below the button is an 'OR' separator and a 'Create New Session' button.

SignBridge

Home About Help Contact

userone

Join a Session

Enter your details to start the call.

Your Name

Room Code

Start Call

OR

Create New Session

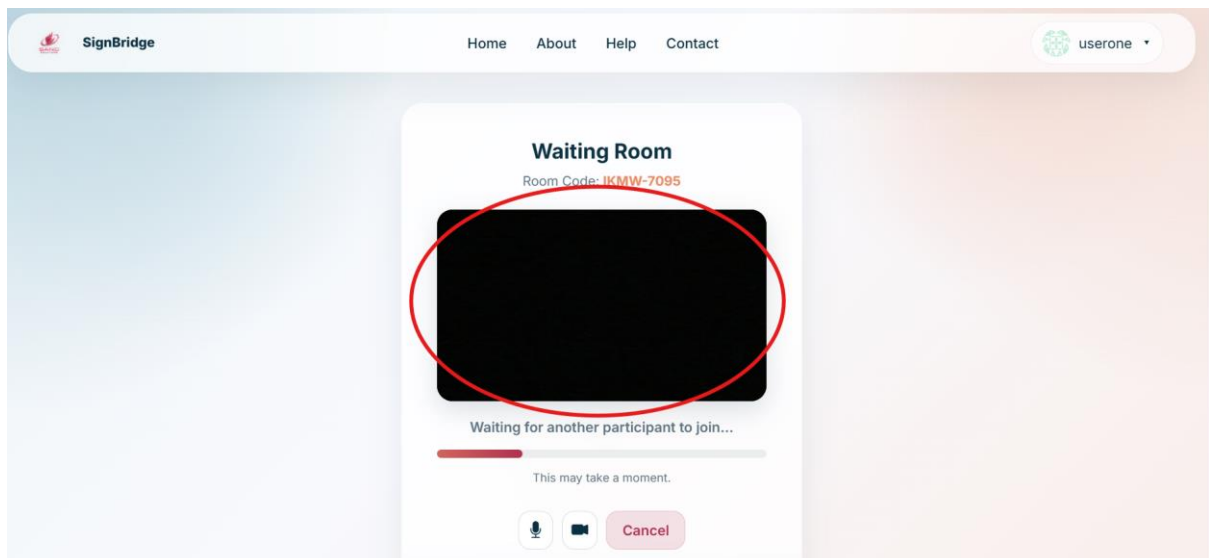
4. You will be taken to the video call interface.

4.3 Waiting Room

The waiting room appears after creating or joining a call before the other participant has arrived.

You will See:

Local Video Preview: Your own webcam feed so you can check your positioning and lighting.



The image shows the 'Waiting Room' interface on the SignBridge website. It features a large black rectangular video preview area, which is circled in red. Above the video area, the text 'Waiting Room' is displayed, followed by 'Room Code: IKMW-7095'. Below the video area, there is a progress bar and the text 'Waiting for another participant to join...'. At the bottom, there are icons for microphone and video, and a 'Cancel' button.

SignBridge

Home About Help Contact

userone

Waiting Room

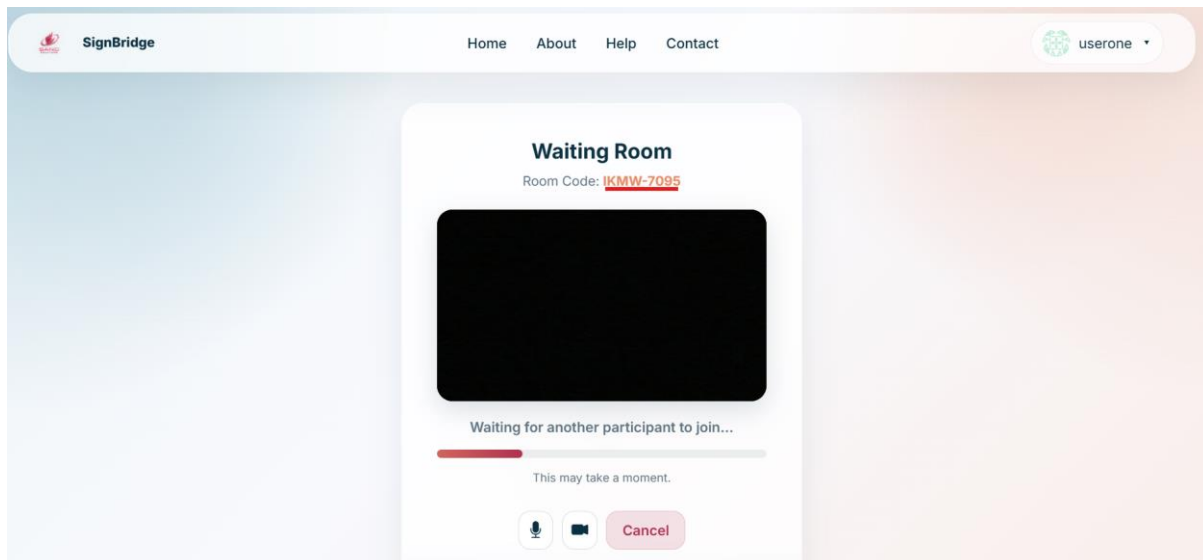
Room Code: IKMW-7095

Waiting for another participant to join...

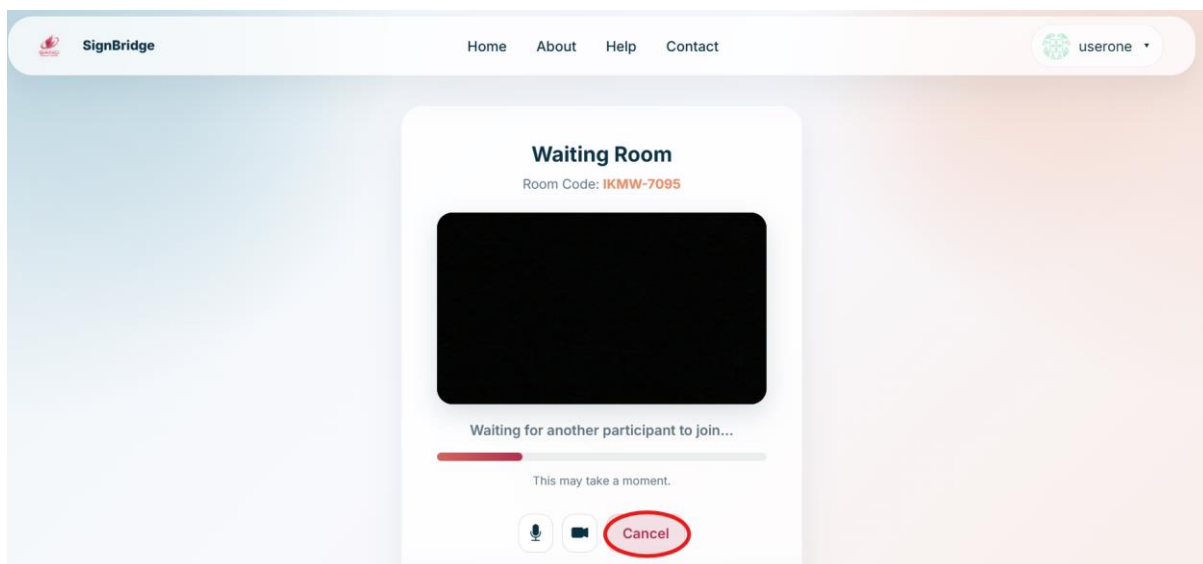
This may take a moment.

Cancel

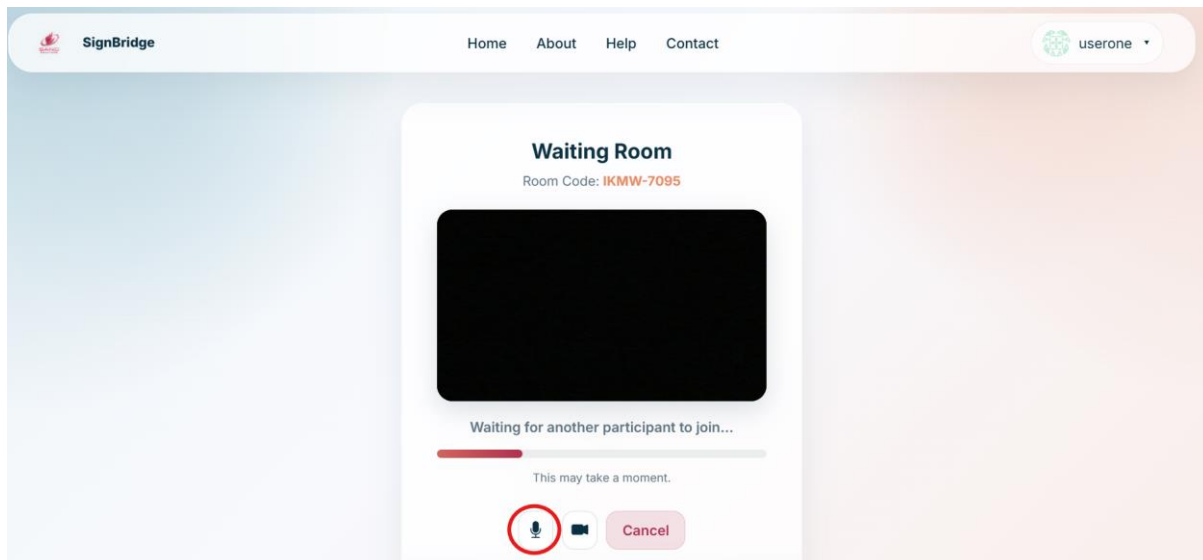
Room Code Display: The Room Code for the current session.



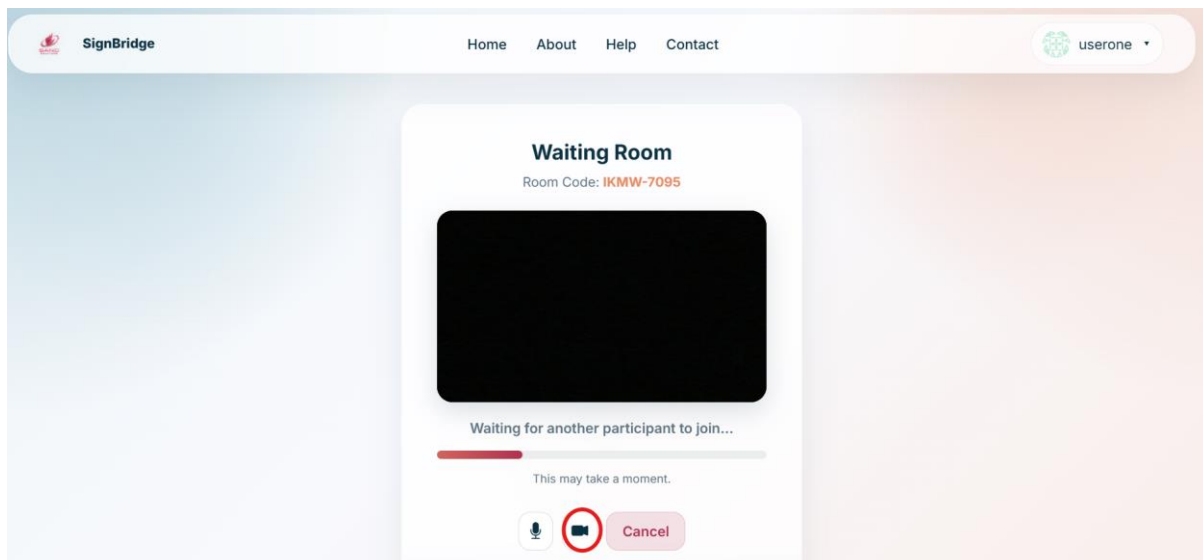
Cancel Button: Exit the waiting room and return to the dashboard.



Mute/Unmute: Control your microphone permissions.



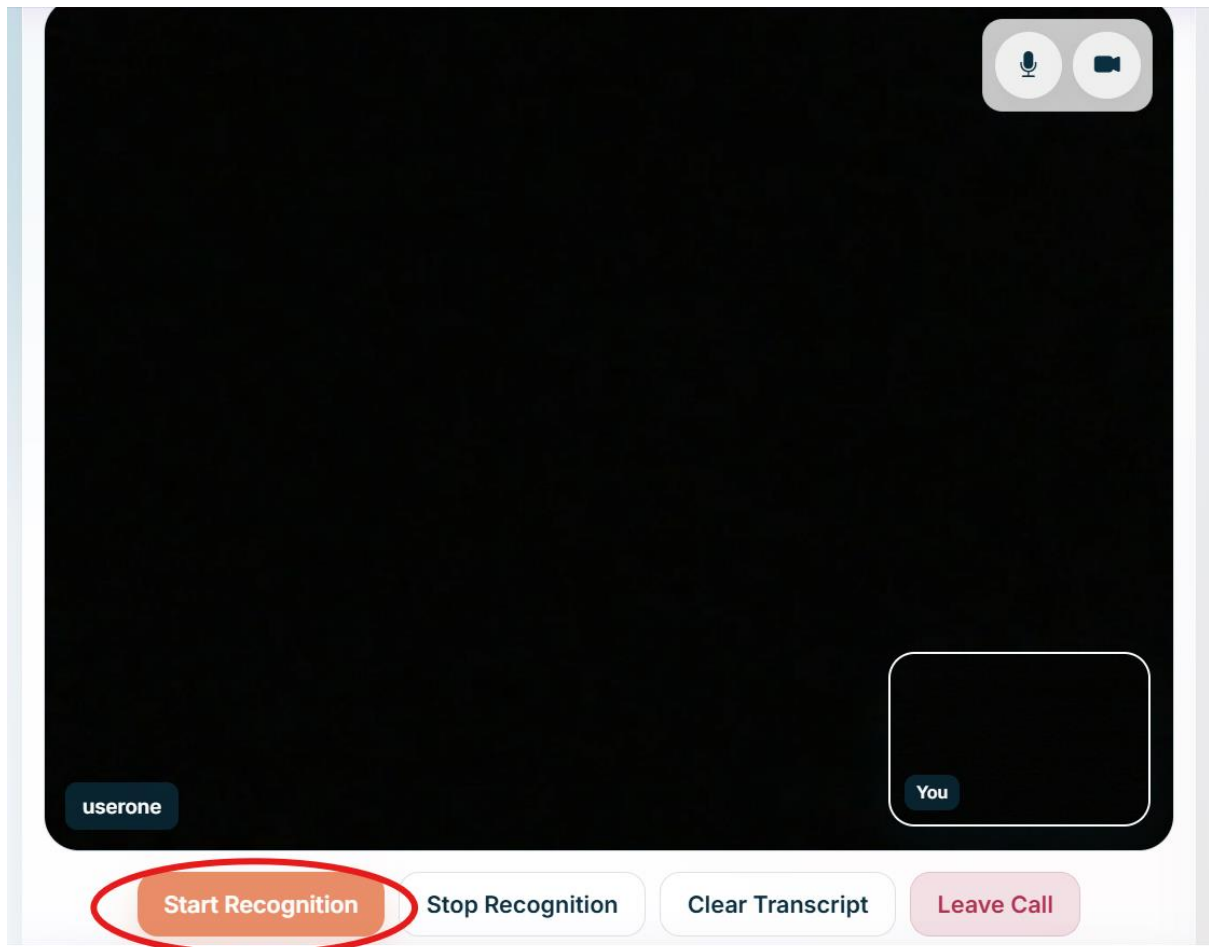
Camera On/Off: Control your camera permissions.



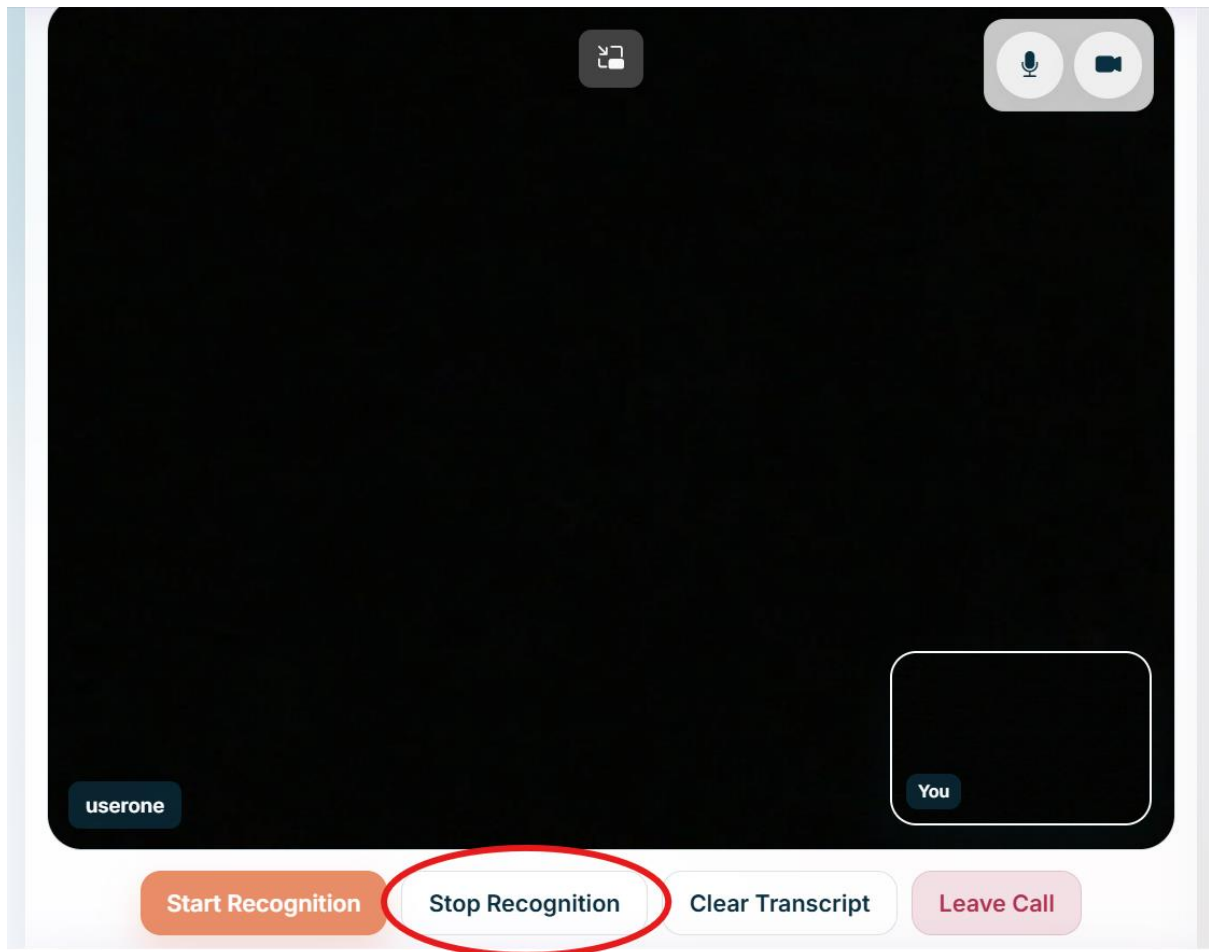
4.4 Using Gesture Recognition

This is the core feature of SignBridge. The system detects your hand gestures and translates them into text.

1. Click on “Start Recognition” to start the gesture recognition process.



2. Raise your hands towards the camera and make a fingerspelling gesture.
3. The system detects your hand.
4. A 5 second timer will run as the model recognises the letter.
5. Click "Stop Recognition" to end the gesture recognition process.

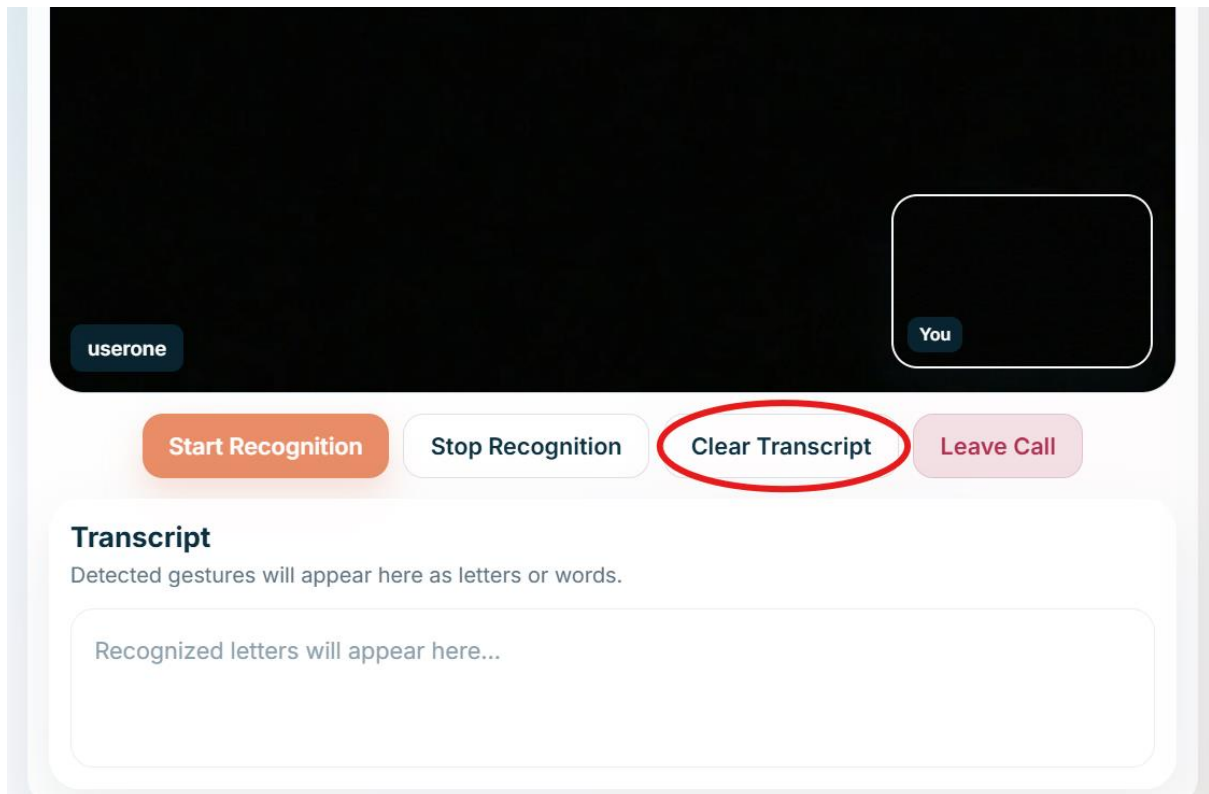


4.5 Transcript

The transcript is a running log of all recognised letters that appear during the call.

Clearing the Transcript

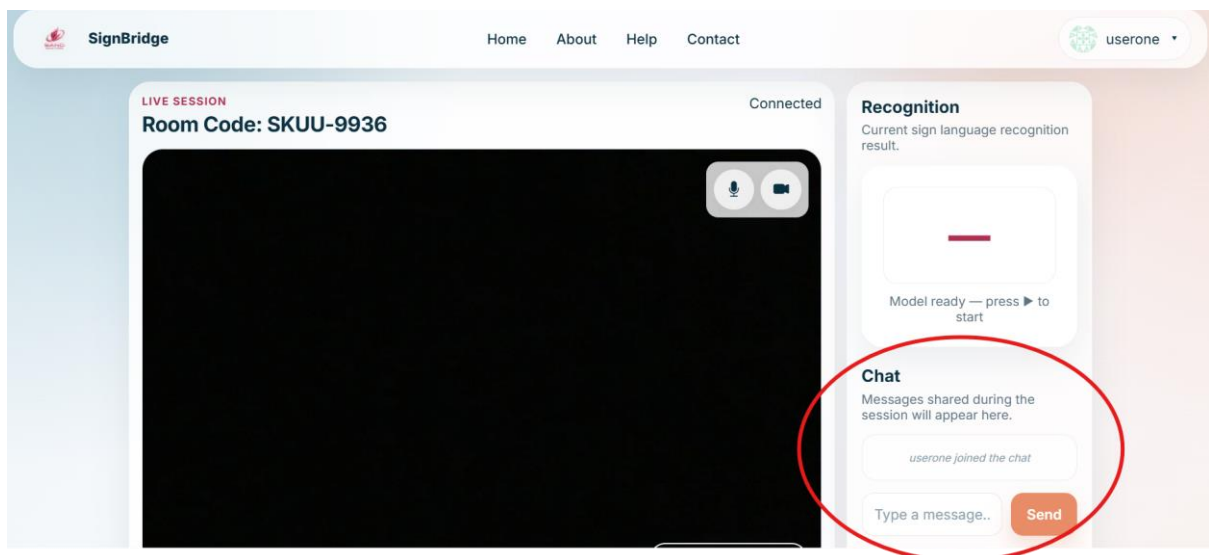
1. Click the “Clear Transcript” button below the video call panel.



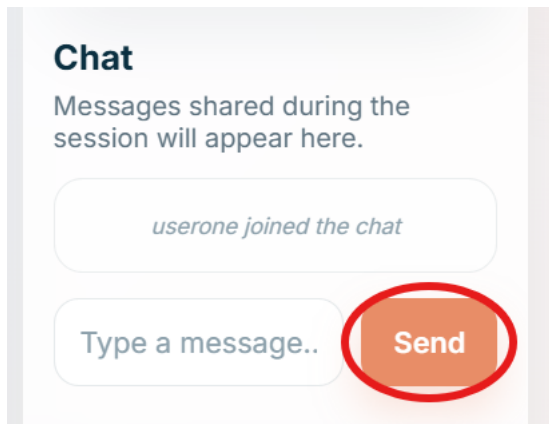
2. All text in the transcript will be cleared.

4.6 Using the Chat

1. The chat box is on the right side of the screen.

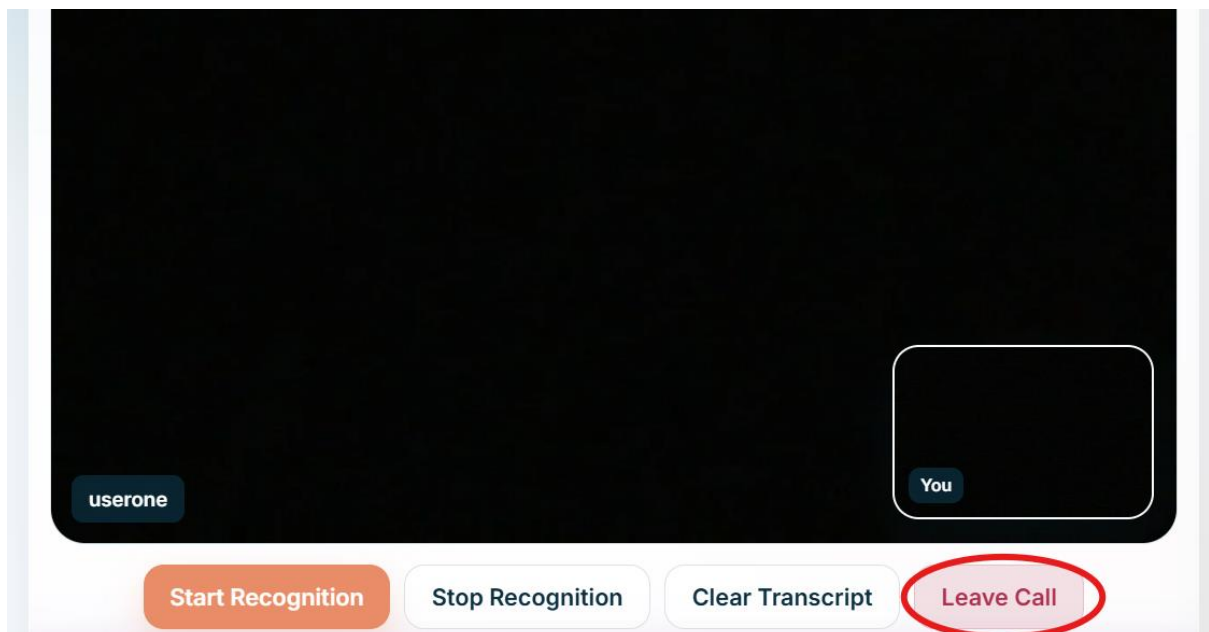


2. You can type messages and click “send”.



4.7 Ending a Call

1. Click the “Leave Call” button.



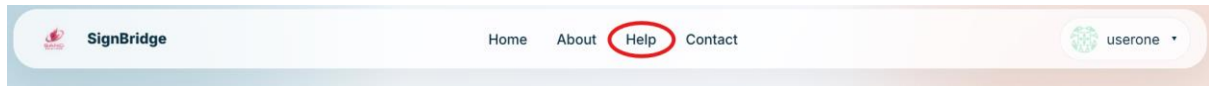
2. Confirm if the prompted message pops up.
3. You will be redirected to the Dashboard.

5.0 Help and Support Features

SignBridge includes several resources to help you. This section explains how to access and use these features.

5.1 Help Section

From the dashboard, click the “Help” button to open the Help Section.

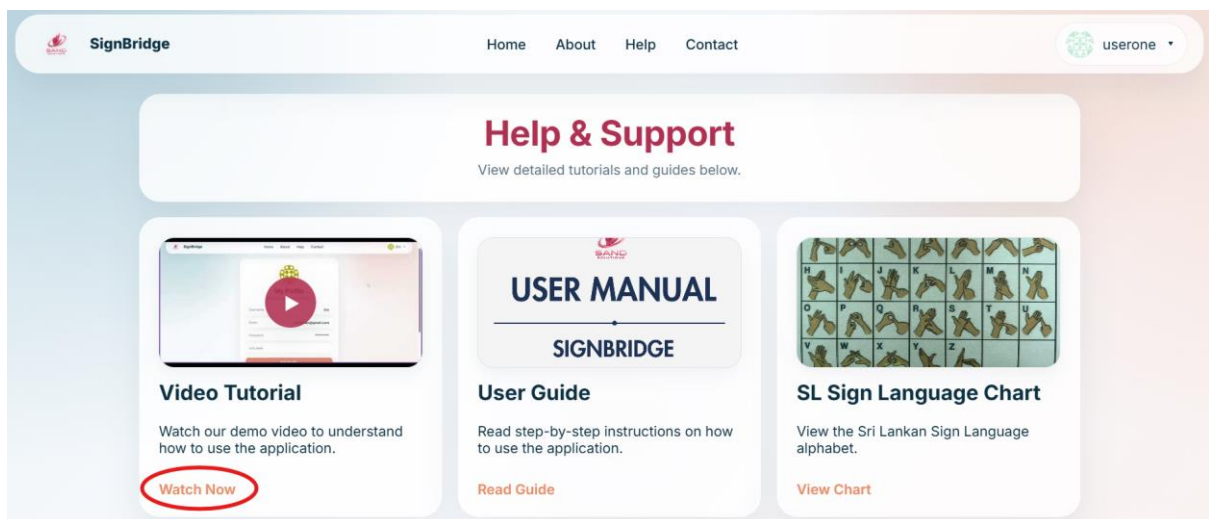


The Help Section contains the following resources:

Resource	Description
Video Tutorial	A short video demonstrating how to use SignBridge.
User Guide	A downloadable PDF version guiding you how to use SignBridge.
SLSL Reference Chart	Visual chart of the supported sign language letters.

5.2 Video Demo

Click on “Watch Now” in the Help Section to be directed to a page where you can play the video.



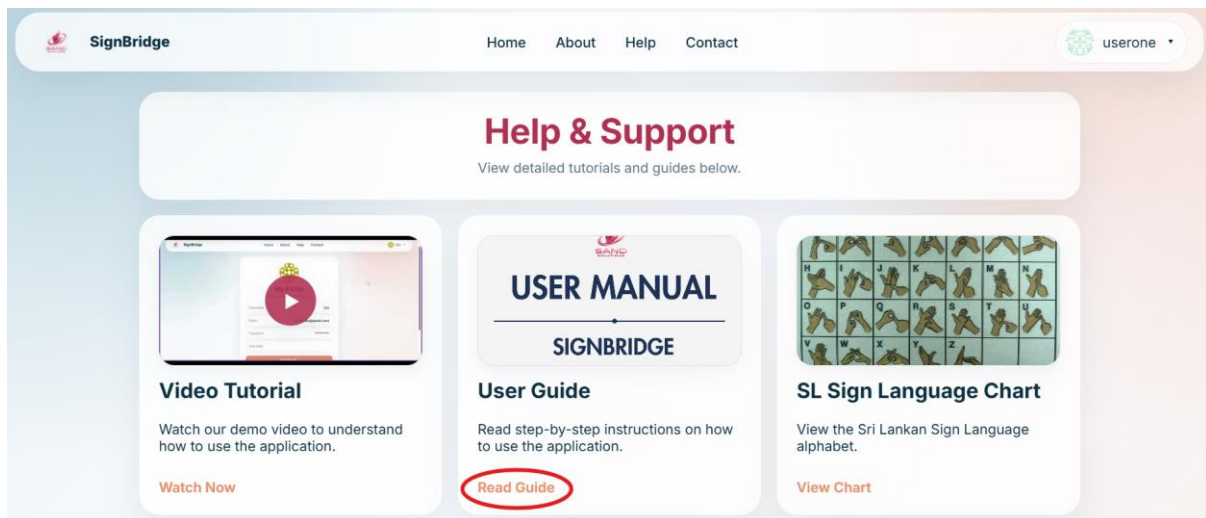
What the Video Demo Covers:

Topic	Description
Getting Started	Creating an account and logging in.
Making a Call	Creating and joining a session.
Hand Positioning	How to position your hands for best recognition.
Using Recognition	How gestures are translated to text.

Call Controls	Using mute, camera and end call features.
---------------	---

5.3 User Guide

Click on “Read Guide” in the Help Section to download the pdf.

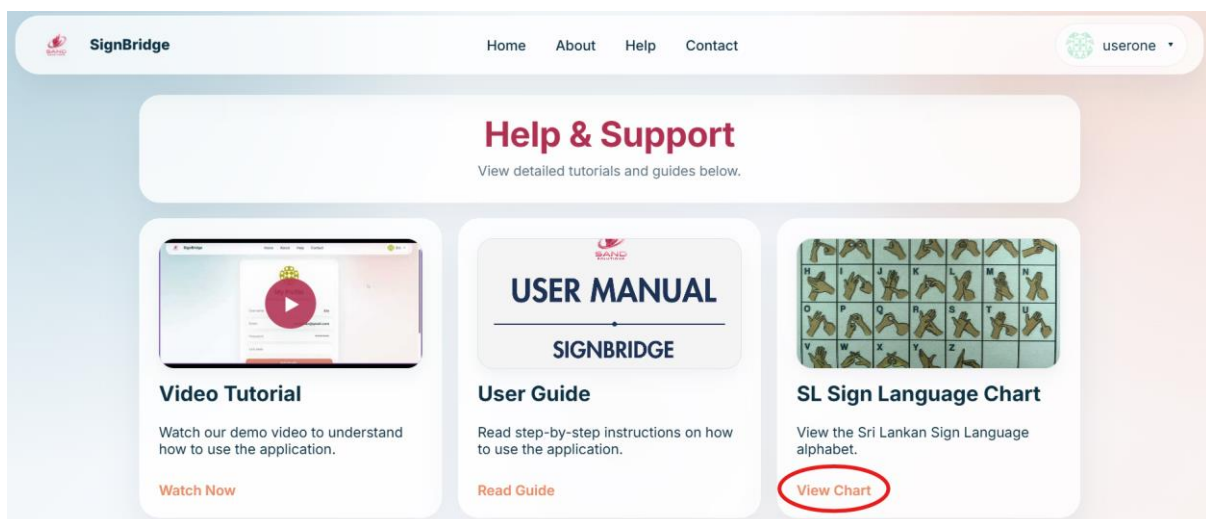


What the User Guide covers:

Topic	Description
Getting Started	Creating an account and logging in.
Making a Call	Creating and joining a session.
Hand Positioning	How to position your hands for best recognition.
Using Recognition	How gestures are translated to text.
Call Controls	Using mute, camera and end call features.

5.4 SLSL Reference Chart

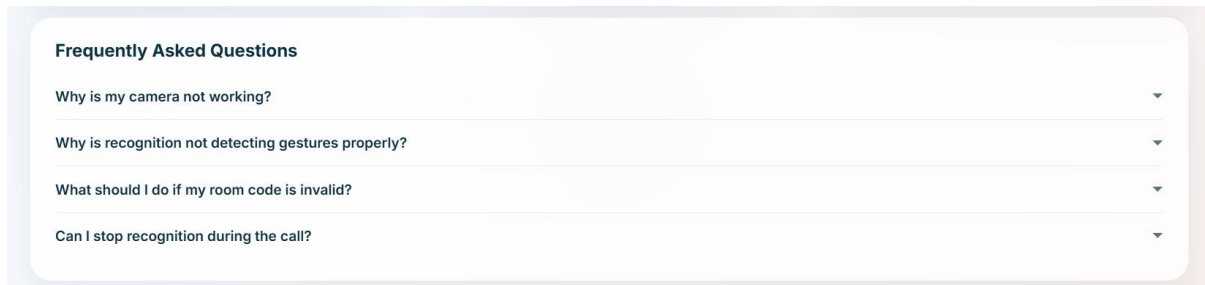
Click on “View Chart” in the Help Section to be directed to a page with the chart.



***Note:** If you are a hearing user, familiarize yourself with the correct hand shapes and practice forming each letter correctly.*

5.5 Frequently Asked Questions

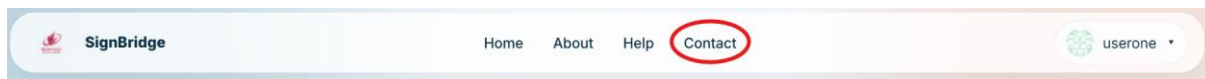
A FAQ is available at the bottom of the help page for you to toggle through common questions.



5.6 Contact Us

If you require further assistance not covered by the Help Section resources, reach out to the support team via the Contact page.

1. From the Dashboard, navigate to the “Contact” page.



2. Fill in the contact form with your name, email address, subject and message.

3. Click “Send Message”.

Contact Us

Have questions or need assistance? We're here to help.



Email Us

admin.signbridge+inquiries@gmail.com



Call Us

+94 71 436 0180



Working Hours

Mon-Fri: 9:00 AM - 5:00 PM

Contact Form

Fill in the form below and we'll get back to you.

Your Name

Your Email

Subject

Message

Send Message

6.0 Error Handling/Troubleshooting

This section covers common issues and how to fix them.

6.1 Camera Not Working

Issue	Solution
No Video Feed	Click “Allow” when your browser asks for camera permission.
Camera permission denied	Click the camera icon in your browser address bar and change to “Allow”.

6.2 Gesture Not Recognised

Issue	Solution
No letters appear	Ensure you are in a well-lit room with light facing you.
“No Hand Detected” message appears	Position your hand in the centre of the frame a few feet from the camera.
Gesture recognition is poor	Avoid wearing long sleeves or jewellery that cover your hands and fingers.
Recognition not starting	Refresh the page and check your internet connection.

6.3 Cannot Join a Call

Issue	Solution
“Room not found. Check the code and try again.” message appears	Check the Room Code for any typos.
Call not found	The call may have ended. Ask the host to start a new session or for the new Room ID.

6.4 Poor Video Quality/Lag

Issue	Solution
Blurry or frozen video	Close other applications on your browser that’s using the internet.
Delay in recognition	Move closer to your router or use a wired connection.

7.0 Current Limitations

This section covers what the application cannot currently do.

7.1 Recognition Scope

Limitation	Explanation
Alphabet only Recognition	The system recognises the static Sri Lankan Sign Language letters only. It does not recognise full words, phrases or dynamic signs.
Limited Set of Letters	The current version only supports 10 alphabet letters (A, E, I, O, U, L, N, R, S, T)
Static Gestures Only	Moving gestures such as waving or signing whole words are not recognised. Each sign must be signed clearly and held for a few seconds.
One Signer	The system tracks only one pair of hands at a time. It will not focus on multiple hands within the frame.

7.2 Environmental Factors

Limitation	Explanation
Lighting Dependent	Recognition works best in well-lit room. Other environments have a possibility of reducing accuracy.
Camera Position	Your hands must be clearly visible and centred in the frame. If not, the recognition stops.
Background Clutter	Busy or moving backgrounds may interfere with accurate hand detection.

7.3 Technical Limitations

Limitation	Explanation
One to One Calls Only	Group calls with more than two participants are not supported in this version.
Web Only	The application works in a web browser only. There are no native mobile or desktop apps currently.
Internet Required	An active internet connection is required to make calls. Offline mode is not available.
Browser Dependent	The application works only on supported browsers. See 2.2

7.4 Privacy Limitations

Limitation	Explanation
Transcript not Saved	Transcripts are not saved after a call ends.
No Call Recording	Calls cannot be recorded within the application.

8.0 Privacy and Data Handling

Feature	Description
Local Processing	All video frames are processed on your device. No video data is uploaded to any server.
No Permanent Storage	Session data such as transcripts and video frames are deleted once the session ends.
User Control	Users can manually delete session data at any time.
Secure Authentication	Passwords are hashed and stored securely.